Komal Rani

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PROFESSIONAL SUMMARY

Result-driven and highly skilled **Senior Associate Consultant** with **5 years** of comprehensive experience in backend software development, specialising in **Java** ecosystem architectures. Proven track record in designing scalable microservices, executing flawless migrations, and integrating robust security layers. Expert at enhancing continuous integration pipelines and leveraging cloud infrastructure to deliver high-performance enterprise applications.

CORE COMPETENCIES

- **Programming:** Java 8 / 11 / 17
- **Backend Frameworks:** Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Spring Security
- **Architecture & APIs:** Microservices, REST API Design, API Gateway, Swagger
- **Databases & Caching:** MySQL, MongoDB, Redis, SQL, JPQL
- **Messaging & Resilience:** Apache Kafka, JMS, Circuit Breakers (Resilience4j)
- **Cloud & DevOps:** AWS (EC2, RDS, IAM, CloudWatch), Docker, Kubernetes, Jenkins, CI/CD, Maven
- **Testing & Quality:** JUnit, Mockito, Postman, SonarQube

PROFESSIONAL EXPERIENCE

Senior Associate Consultant — Infosys Ltd., Pune *Dec 2024 – Present*

- **Architecture Upgrade:** Spearheaded the comprehensive migration and structural upgrade of enterprise-grade applications from **Java 8/11 to Java 17**, and **Spring Boot 2.7.x to 3.x** architectures.
- **Layered Development:** Built scalable, secure controller-service-repository backend structures implementing robust global exception handling, data validation, and automated transactional management.
- **Security Integration:** Designed inter-service communication protocols using **JWT and OAuth 2.0** frameworks to standardise identity access management and application security.
- **Distributed Resilience:** Architected high-availability data pipelines utilising **Apache Kafka** and implemented event-driven patterns (Saga orchestrations), circuit breakers, and fallback mechanisms to ensure **100% eventual consistency**.
- **Infrastructure Management:** Managed cloud runtime operations utilising **Redis** caching mechanisms, **Spring Boot Actuator** health states, centralized log tracking, and distributed tracing across environments.
- **Database Tuning:** Optimized complex data distribution architectures through advanced index configurations, proactive fetch strategies, and automated data pipelines deployed on **AWS** ecosystems.

Software Engineer — Capgemini, Chennai *Nov 2021 – Dec 2024*

- **Modernization:** Modernized legacy production applications by refactoring synchronous code blocks into functional **Java Streams, Lambdas**, and modular asynchronous components.
- **Data Access Layer:** Designed relational database objects and structured optimized multi-table CRUD operations using custom execution plans.
- **DevOps Pipelines:** Automated standard image compilation processes using containerized configurations via **Docker** environments to streamline cross-team deployments.
- **Cloud Monitoring:** Provisioned, scaled, and secured container environments on **AWS** using **IAM roles**, system log metrics, and automated **CloudWatch alarms** to maintain zero downtime.
- **Quality Assurance:** Minimized regressions by executing rigorous regression analysis pipelines, performance testing, and full-coverage automated testing using **JUnit and Mockito**.

EDUCATION

Bachelor of Technology (B.Tech) — Kurukshetra University ***Year: 2020***